The Destruction of Social Capital and Family Flight: Evidence from Casino Openings

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Effect of new casino development on Population and Family Dynamics

- Local officials using new casinos to enrich local economies via economic inflows.
  - Evidence of some short term benefits of employment and tax revenues. (Humphreys **and** Marchand, 2013; Wiley **and** Walker, 2011)
  - Significant evidence of long term negative spillovers on the social strata. (Grinols **and** Mustard, 2006)

- In addition, communities may wish to promote population growth.
- Effects of casinos on population and family formation are not well understood.
- Only event analysis studies (Moellman **and** Mitra, 2013)

Possibility of Resort style Casino Gambling in Texas:

- Proposed Sands Casino in Irving, Texas
- Proposed Irving location close to University of Dallas
  - Irving, TX: Pop: 260k, Inner-ring suburb
  - Lower Median Income for DFW (79k), but similar to the rest of Texas/US
  - Highly Diverse: Demographics and Income
- Why Irving?
  - First mover casino, but not a current destination area.
  - Location is a self-contained, large empty plot: Little to no economic spillovers. Akin to a *walled medieval European city*

▸ Open Interactive Map

Related Work: Economic Activity and Family Formation

- Family and community structures build social capital and improve outcomes for children from low-income families (Chetty, 2026)
- Fertility as consumption: Positive income shocks increase fertility (Kearney, 2023; Gallego **and** Lafortune, 2023)
- Fertility as investment: Positive income shocks increase savings; parents may view children as a way to smooth future consumption (Goldin, 2021; Doepke, 2023)

Related Previous Work: Casinos and Economic and Social Outcomes

- Increased Crime: Casinos caused increases in aggravated assault, rape, robbery, larceny, burglary, auto theft (Grinols **and** Mustard, 2006)
  - Magnitude: 8.6% of property crime and 12.6% of violent crime in casino counties attributable to casinos
- Increases in Economic Outcomes
  - Retail Property values (Wiley **and** Walker, 2011)
  - Increases in Hospitality Industry employment (Humphreys **and** Marchand, 2013)
- Unique for this project: no preexisting casino presence

Research Contribution and Preliminary Results

- Novel work on how a new casino entrant affects marriage, fertility, and population composition, setting akin to Sands' entry into Irving
- Unclear direction: Crime increases (-) vs. Employment gains (+)

Initial Results

- Casino $\rightarrow$ County level children population drops
- Casino $\rightarrow$ 2.01 percentage point decline in fertility rates for women, 3.17 drop for men
- Casino $\rightarrow$ drop in School enrollment and school age population, around 6 percentage point drop 3 years after opening

Explanatory variable of interest: 2003-2023 US county-level casino openings

- Created a database of new casino openings by county and adjacent counties from 2003 to 2023 (to match timeframe of ACS data)
- Previous research finds increased economic activity for 1-2 yearsbefore the casino opening.
  - Time of treatment effect is two years before casino opening.
  - Event Study Analysis: Base outcome is the 3rd lag before opening.

Treatment: New casino counties. Only includes new casino counties in the data time frame, no previous casinos or additional casinos. (19 total)

Control: No Casino or casino adjacent counties or previous existing nearby casino to prevent spillovers clouding effects (Akin to a Donut Diff in Diff design.)

2003-2023 US county-level casino openings

casino_name	tuyear	Location
Century Casino Cape Girardeau	2012	Cape Girardeau, MO
Chukchansi Gold Resort & Casino	2003	Coarsegold, CA
Diamond Jo Casino - Worth	2006	Worth County, Iowa
FireKeepers Casino Hotel	2009	Emmett Charter Township
French Lick Resort	2008	French Lick, IN
Harlow's Casino Resort & Hotel	2007	Refuge, MS
Hollywood Casino Columbus	2012	Columbus, OH
Horseshoe Indianapolis	2009	Shelbyville, IN
Ilani Casino Resort	2017	Ridgefield, WA
Isle Casino Waterloo	2007	Waterloo, IA
Kansas Crossing Casino	2017	Pittsburg, KS
Kansas Star Casino	2011	Mulvane, KS
Ocean Downs	2011	Berlin, MD
Oxford Casino	2012	Oxford, ME
Rivers Casino Portsmouth	2023	Portsmouth, VA
Saganing Eagles Landing Casino	2007	Standish, MI
Seminole Hard Rock Hotel & Casino Holloywc	2004	Hollywood, FL
Seminole Hard Rock Hotel & Casino Tampa	2003	Tampa, FL
Yellow Brick Road Casino	2015	Chittenango, NY

Dependent Variables: population and family formation measures

- ACS micro data from 2000-2023
- Adult and Child Population data
  - Total, Child Rearing (18 to 40), Children ($\leq$ 17), School age (5-17)
  - Residence duration: Broken down by gender
- Family and fertility data:
  - marital status (base: not married currently)
  - has children
  - number of children
- School age populations and enrollments
- To tease out treatment effects, we use two methods: Event Study and Staggered Difference and Differences (DiD).

Treatment vs. Control county level differences

- Notable aggregate differences between treatment casino counties vs. control counties
  - 32.0% vs 38.0% married, 36.2% vs. 41.0% has children, 71 vs. 82 children per 100 ppl.
  - 63.1% vs 70.8% White
  - Treated counties skew more rural than control counties.
- Gender and Age are similar

Method 1: Event Study

- Compares outcomes *before vs. after* an event (e.g., casino opening) using never-treated places as a benchmark.
- Lines up each county by **event time** (years relative to opening), then estimate the average difference at each event year.
- What you look for:
  - No clear pre-trend / "parallel trends".
  - Coefficients show the size and timing of the effect (immediate, gradual, temporary, persistent).

Method 2: Staggered Difference and Differences (DiD)

- Staggered Difference and Differences estimated with two-way fixed effects (FE) specification:

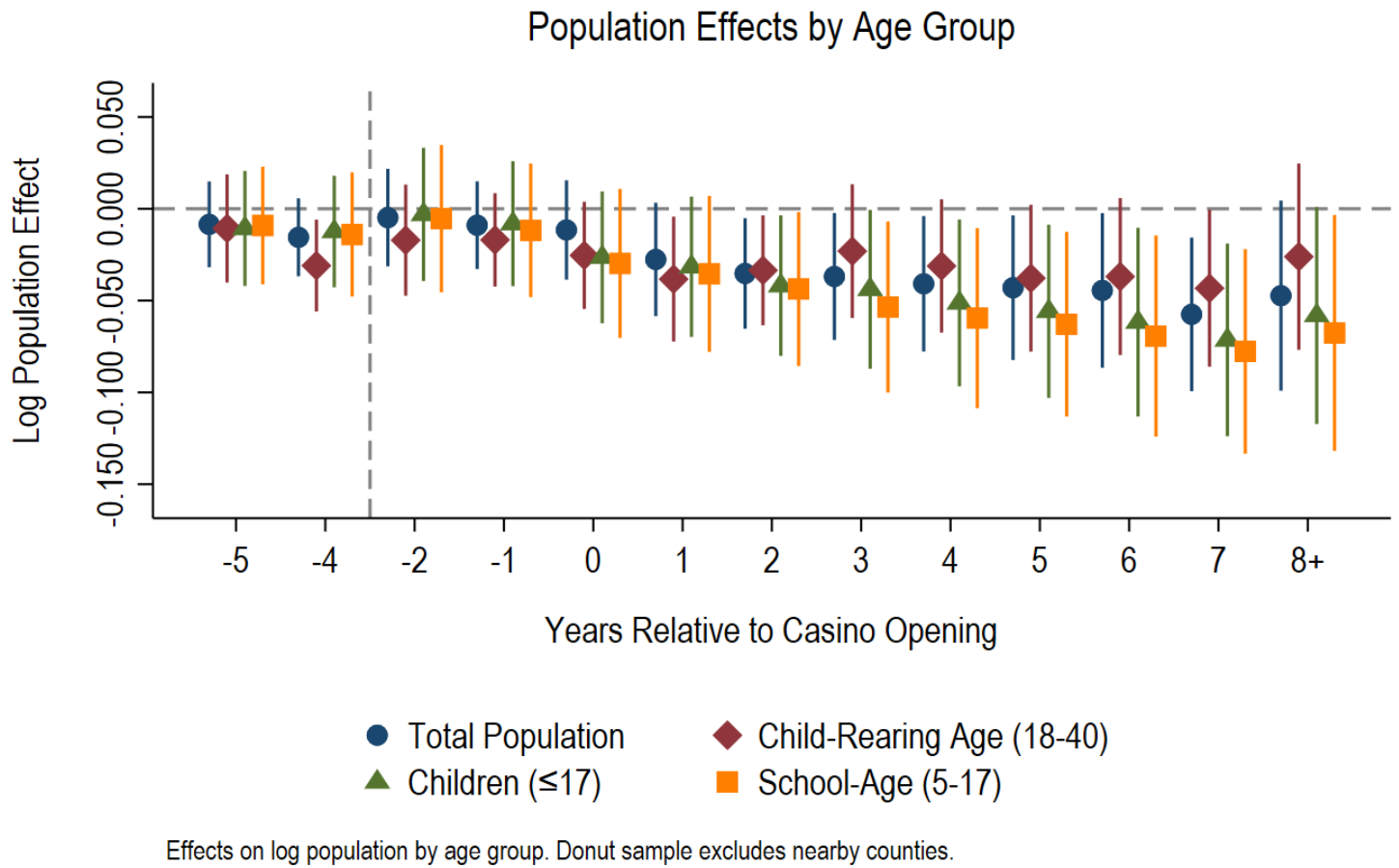
$$Y_{ct} = \alpha + \beta \cdot \text{Post}_{ct} + \mathbf{X}'_{ct}\gamma + \mu_c + \lambda_t + \varepsilon_{ct} \quad (1)$$

  - Y_{ct} is the DV in county c and year t
  - Post_{ct} is an indicator equal to one if county c has an operating casino in year t
  - $\mathbf{X}_{ct}$ includes demographic controls
  - μ_c and λ_t are county and year fixed effects
  - Standard errors clustered at the county level
  - Weighted by overall population (low population county averages do not hold the same weight as densely populated counties)

Staggered Difference and Differences (DiD):

- Causal technique to estimate treatment effects (new casino)
- In many applications, we want to estimate the effect of a policy or economic change
- However, the policy assignment is not necessarily uncorrelated with group characteristics
  - A new casino may be located specifically in treated areas with declining population and family formation
  - DiD uses, as a control, areas that were never treated
- The treatment effect (effect of the casino) is the difference in the DV of the treated areas minus the difference in the DV of the control areas.
- Fixed Effects estimation accounts for county and time-specific effects that may confound results

Results for log population by age group



Effects on log population by age group. Donut sample excludes nearby counties.

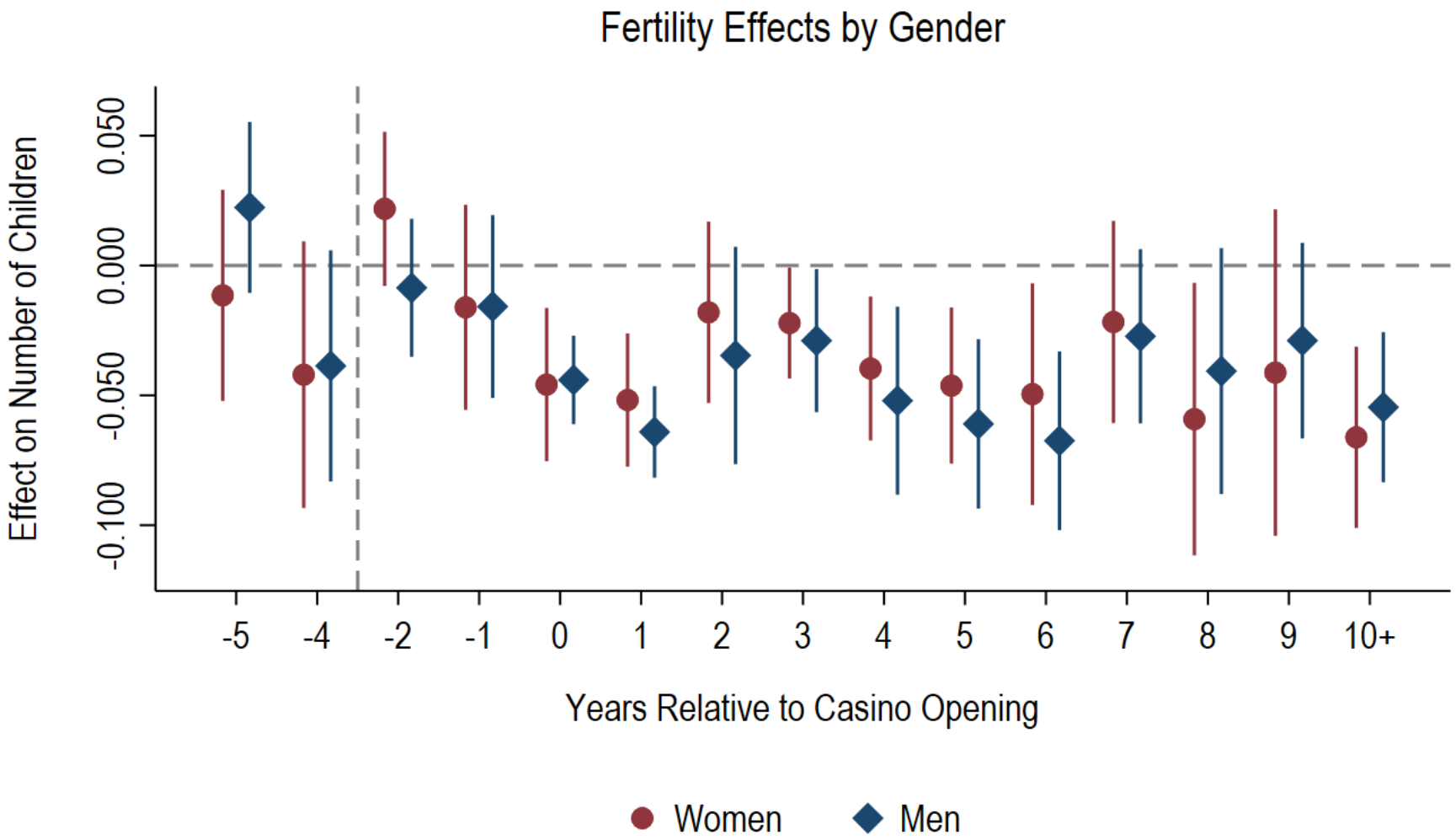
Stronger effects on children relative to adults suggest family formation vs. general population decline

Diff in Diff results on Fertility

	Women			Men		
	(1) Marriage	(2) Child	(3) Num Kids	(4) Marriage	(5) Child	(6) Num Kids
Back2	0.0077	-0.0049	-0.0201**	-0.0007	-0.0104**	-0.0317***
	(0.0050)	(0.0056)	(0.0084)	(0.0052)	(0.0051)	(0.0079)
Obs	7,414,164	7,414,164	7,414,164	7,506,358	7,506,358	7,506,358
R-sq	0.208	0.272	0.243	0.220	0.229	0.206

Standard errors in parentheses
All regressions include county and year fixed effects, age and age squared.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility: Event-Study Plot



Event study effects on number of children. Donut sample excludes nearby counties.

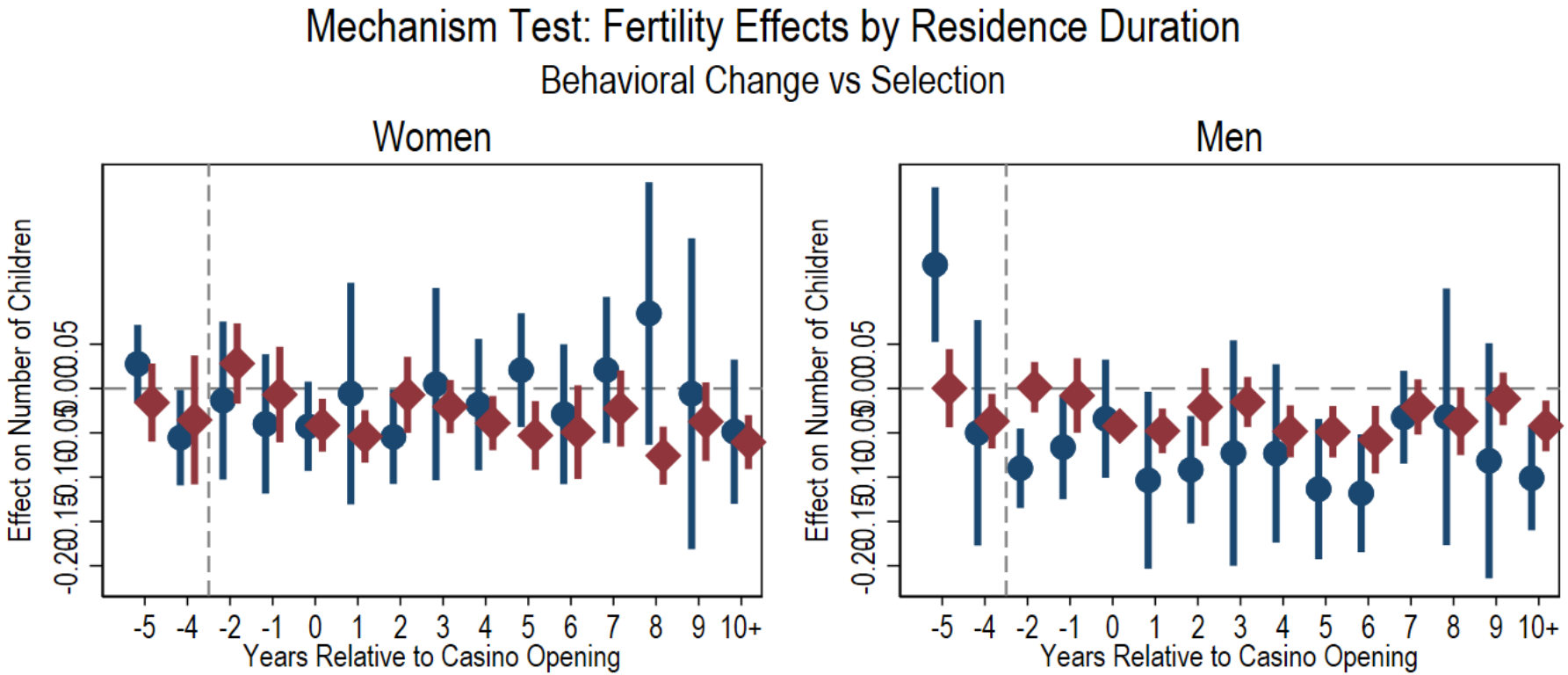
Fertility Mechanism: DiD Effects by Residence Duration

Table: Family Formation Effects by Gender

	Women		Men	
	(1) ST Women	(2) LT Women	(3) ST Men	(4) LT Men
post_treatment_back2	-0.0241***	-0.0054	-0.0238***	-0.0864***
	(0.0063)	(0.0203)	(0.0081)	(0.0163)
Observations	4,696,769	1,080,303	4,875,631	994,905
R-squared	0.235	0.135	0.174	0.151

Long Term (5+), Short Term (-5)
Standard errors clustered at county level in parentheses.
All regressions include county and year fixed effects, age and age squared.
Donut sample excludes nearby counties.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility Mechanism: Event-Study Plots



Drop in fertility for long-term male residents at casino announcement.

School Effects

Table: Event-Study: School-Age Population & Enrollment (Log DV)

Event Time	School-Age Pop	Total Enroll.	Public Enroll.
t-5	-0.025*	-0.025*	-0.020
t-4	-0.030	-0.024	-0.025
t-2	-0.009	-0.004	0.005
t-1	-0.014	-0.010	-0.021
t	-0.029	-0.023	-0.027
t+1	-0.039**	-0.032	-0.022
t+2	-0.049**	-0.045**	-0.031
t+3	-0.063***	-0.058***	-0.043*
t+4	-0.067***	-0.066***	-0.045*
t+5	-0.056**	-0.054**	-0.036
t+6	-0.068***	-0.061**	-0.035
t+7	-0.076***	-0.070***	-0.058**
t+8+	-0.067**	-0.064**	-0.044

Stars indicate significance; SEs omitted for clarity.
All variables in logs. School-age = ages 5–17. Donut sample excludes nearby counties.

Next Steps

- Needs a more robust parallel trends analysis, including placebo tests.
- Need to employ staggered Diff in Diff corrections, specifically Calloway and Sant'Anna method.
  - Propensity score matching to create a control group that is more similar to the treatment group.
  - A key concern given the "ruralness" of the treatment counties.

Thank you.



Click to open the interactive map: ▸ Open Interactive Map

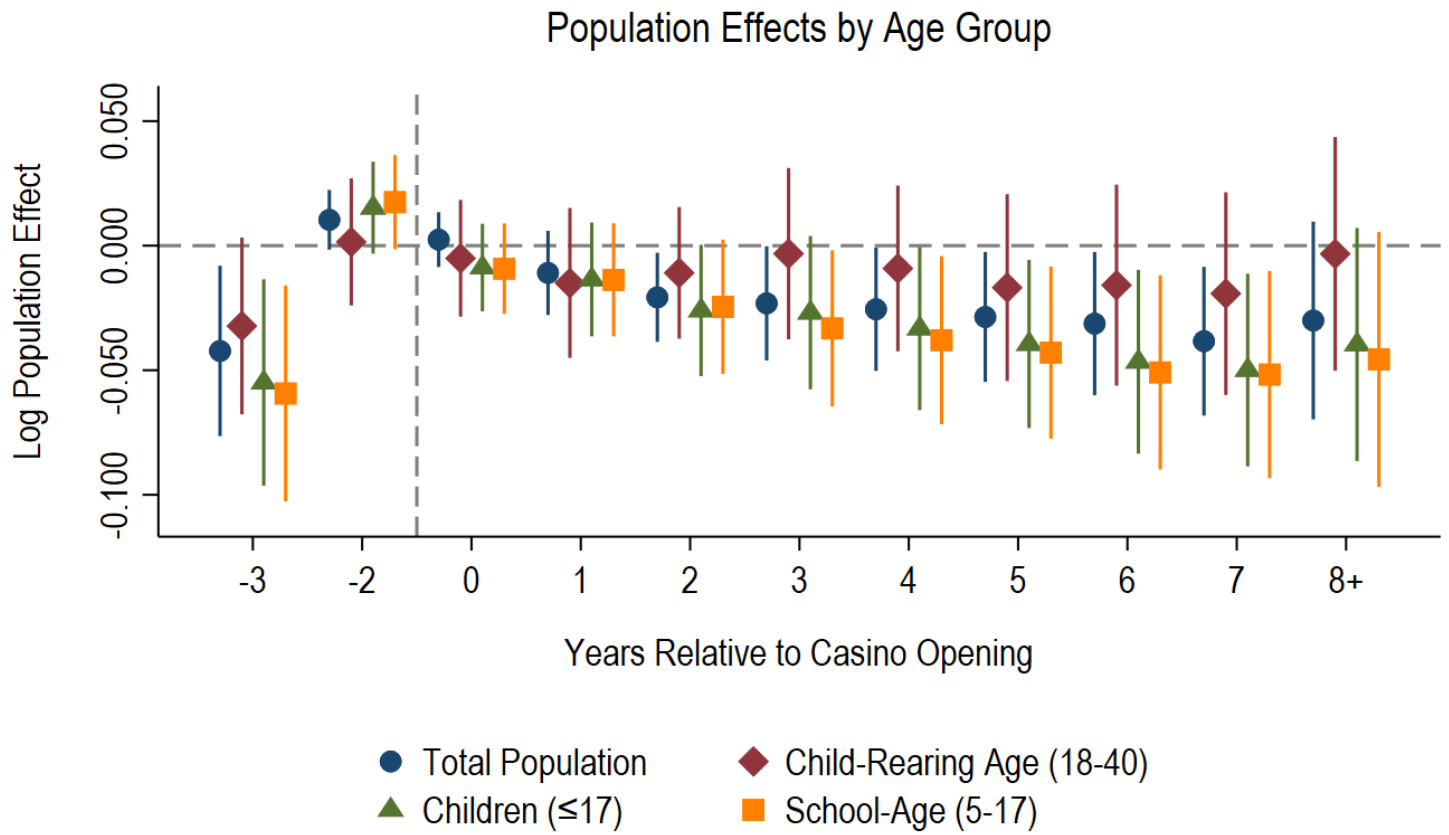
Summary Stats

Table: Summary Statistics by County Type

	(1) Casino Counties		(2) Nearby Counties		(3) Control Counties	
	mean	sd	mean	sd	mean	sd
married	0.320	0.467	0.332	0.471	0.380	0.485
has_children	0.362	0.481	0.381	0.486	0.410	0.492
num_children	0.707	1.120	0.771	1.178	0.824	1.185
female	0.503	0.500	0.493	0.500	0.495	0.500
age	29.166	6.535	29.050	6.648	29.046	6.686
white	0.631	0.483	0.682	0.466	0.708	0.454
black	0.215	0.411	0.138	0.345	0.131	0.338
hispanic	0.203	0.402	0.233	0.422	0.175	0.380
N	391117		486467		15972212	

Notes: Individual-level statistics weighted by person weights.
Sample includes individuals aged 18-40.

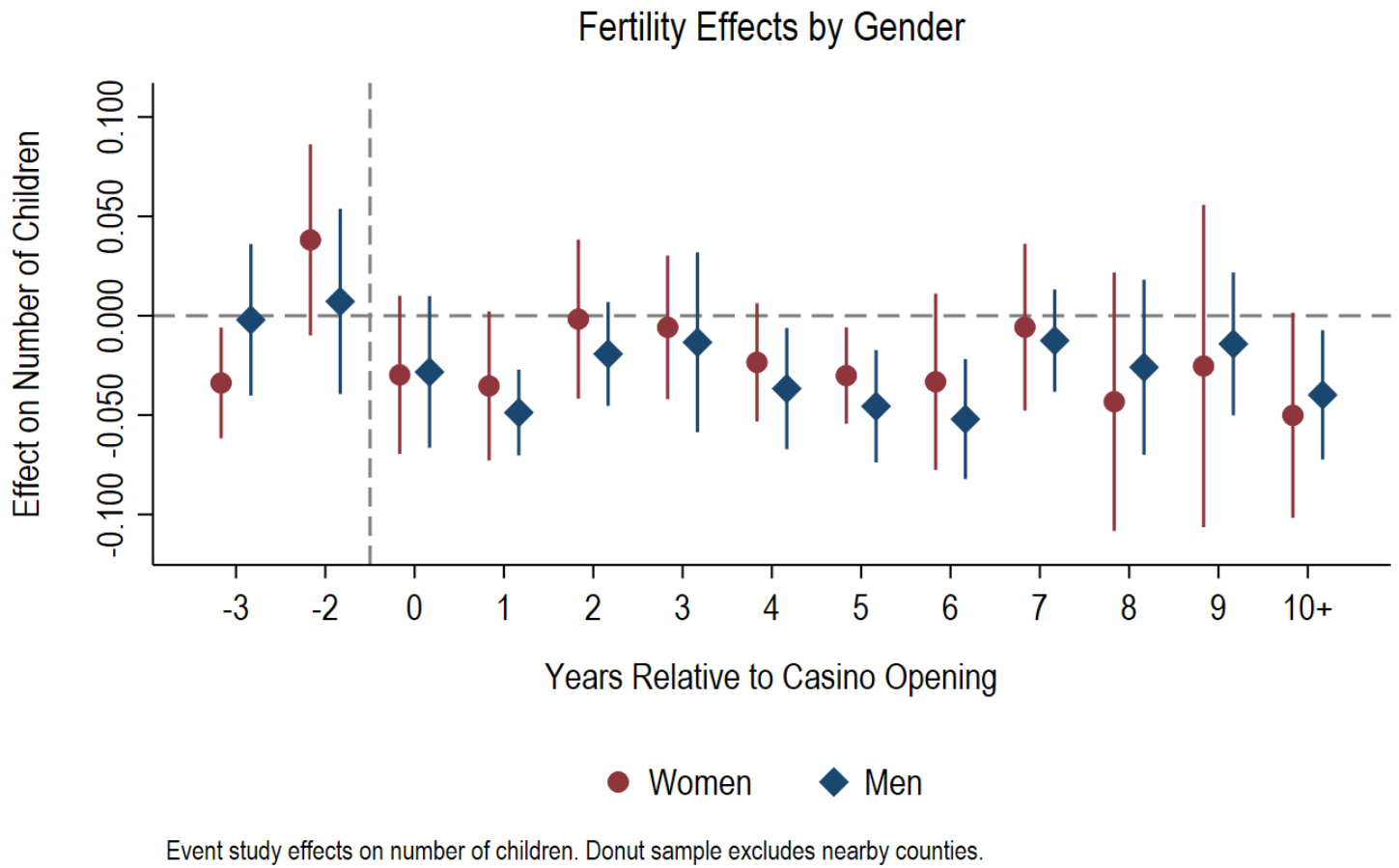
Results for log population by age group (standard Event Study)



Effects on log population by age group. Donut sample excludes nearby counties.

Stronger effects on children relative to adults suggest family formation vs.

Fertility



Event study effects on number of children. Donut sample excludes nearby counties.

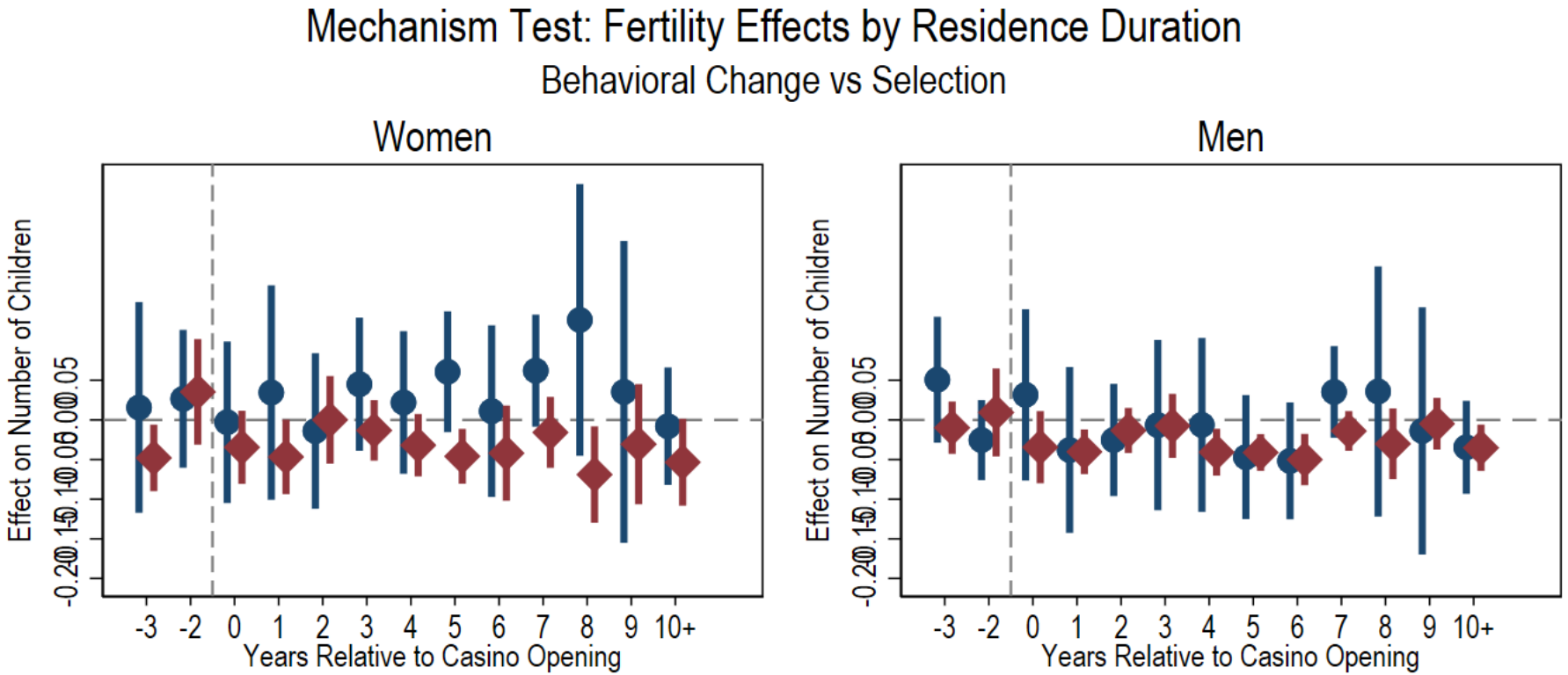
Fertility effects emerge immediately after casino opening and persist throughout our observation window.

Diff and Diff results on Fertility (Standard Treatment timing)

	Women			Men		
	(1) Marriage	(2) Child	(3) Num Kids	(4) Marriage	(5) Child	(6) Num Kids
post_treat	0.0046	-0.0128***	-0.0551***	0.0048	-0.0045	-0.0262***
	(0.0041)	(0.0043)	(0.0138)	(0.0032)	(0.0033)	(0.0089)
Obser	9,135,596	9,135,596	9,135,596	9,230,246	9,230,246	9,230,246
R-squared	0.206	0.267	0.239	0.218	0.225	0.203

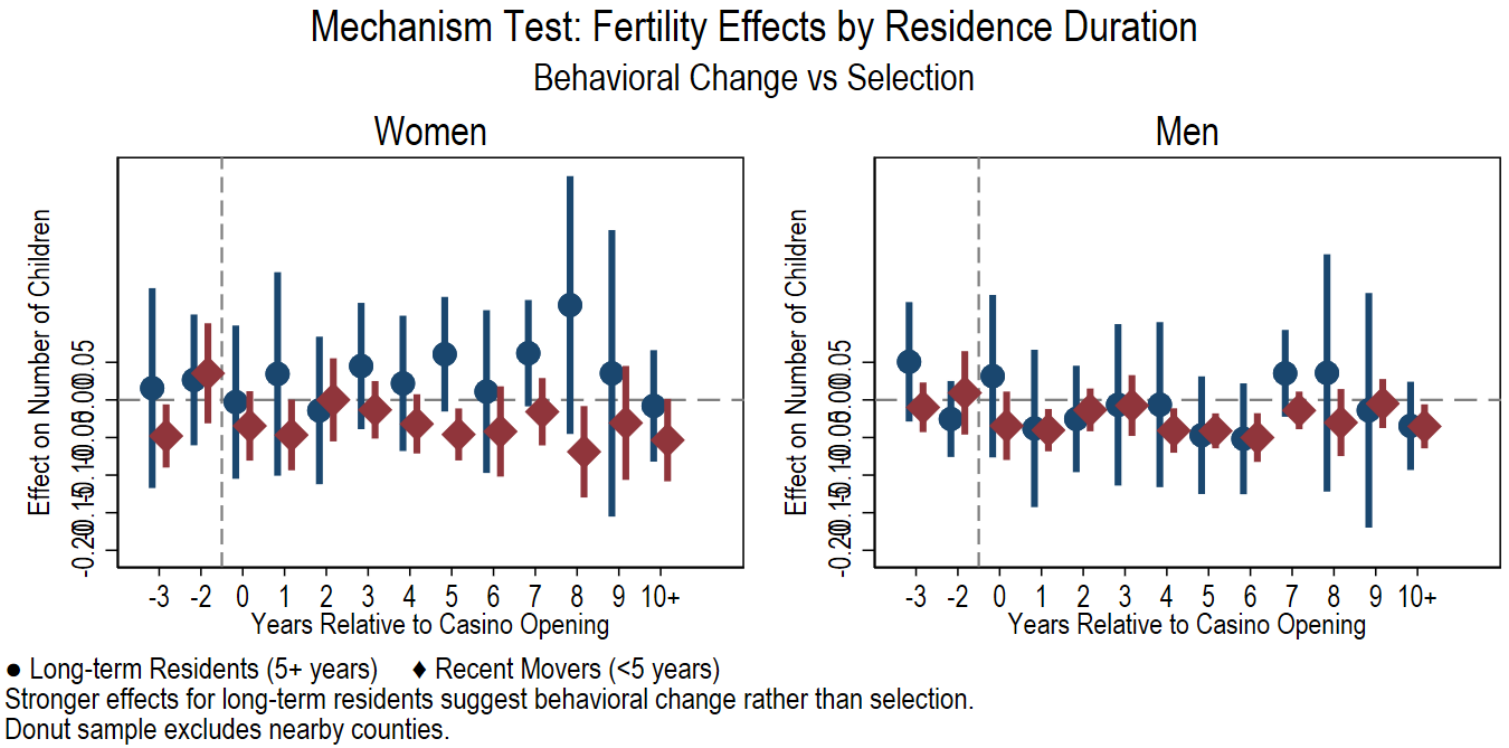
Standard errors in parentheses
All regressions include county and year fixed effects, age and age^2
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility Mechanism (Standard Treatment timing)



Possible drop in fertility for female long-term residents at casino opening.

Fertility Mechanism



Drop in fertility for long-term residents at casino opening. Suggests selective out-migration of family-oriented long-term residents rather than behavioral change.

School Pop & Enrollment — Diff in Diff

Table: Percent changes in School populations

	(1) School Age	(2) School Enrolled	(3) Public Enrolled
post_treatment_back2	0.0000	0.0000	0.0000
	(.)	(.)	(.)
Observations	680	680	680
R-squared	0.997	0.997	0.997

Standard errors clustered at county level in parentheses.

All regressions include county and year fixed effects

Donut sample excludes nearby counties. *** p¡0.01, ** p¡0.05, * p¡0.1.

$^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

School Pop & Enrollment — Event-Study Summary

Table: Event-Study: School-Age Population & Enrollment (Log DV)

Event Time	School-Age Pop	Total Enroll.	Public Enroll.
t-3	-0.051***	-0.049**	-0.040**
t-2	-0.006	-0.001	0.009
t	-0.023	-0.019	-0.023
t+1	-0.023	-0.017	-0.004
t+2	-0.039**	-0.034**	-0.020
t+3	-0.046***	-0.042**	-0.026
t+4	-0.054***	-0.052***	-0.028
t+5	-0.054***	-0.052***	-0.031
t+6	-0.062***	-0.056***	-0.028
t+7	-0.069***	-0.064***	-0.047**
t+8+	-0.066**	-0.064**	-0.038

Stars indicate significance; SEs omitted for clarity.

Religious Adherence

Table: Spillover Effects: Casino vs Nearby Counties

	(1) Fertility (W)	(2) Fertility (M)	(3) Child Pop	(4) Religious
Casino County × Post	-0.0433**	-0.0205**	-0.0253	9.8971
	(0.0180)	(0.0086)	(0.0187)	(24.5977)
Nearby County × Post	-0.0256	-0.0106	-0.0158	-42.1267
	(0.0361)	(0.0272)	(0.0289)	(31.4860)
Observations	10381166	10476298	9,160	9,418
R-squared	0.240	0.205	0.992	0.932

Standard errors in parentheses
Tests for spillover effects to nearby counties. Justifies donut design.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Spillover effects on nearby counties

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$^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Casino county fertility declines of 4.33% for women and 2.05% for men. Nearby county effects 2.56% for women and 1.06% for men, though not statistically significant.

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